State Machines – Part 4
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Review

- We have discussed Mealy machine and Moore machine.
- We have also discussed state reduction.

Another aspect in Design

- Recall that our state diagram for state reduction had only alphabets inside the circles. We indicated that we need not worry about numerical values for purposes of state reduction.
- To complete the design of a sequential circuit with logic elements, we will need to assign unique codes to the states.

What does this mean ?

- We have to know how many flip-flops we will have in our design.
- This depends on the codes we assign.
- There are many ways to assign codes.
- These include (classical) binary coding, Gray code assignment and one-hot encoding.
- We will show each of these next.

less flip flops

simplify boolean

good for decoding

Assume 5 states

State	Assignment 1 Binary
a	000
b	001
c	010
d	011
e	100

Another possibility for 5 states

Assignment 2, Gray Code
000
001
011
010
110

Third possibility

Assignment 3, One-Hot
00001
00010
00100
01000
10000

Which one is better ?

- The classical method (direct binary) helps to keep the number of flip-flops small.
- The Gray code method is good when we want to place the Boolean functions involved in a map for simplification (the design procedure itself can terminate in a logic diagram or a list of Boolean functions from which a logic diagram can be obtained).
- The third possibility called one-hot method uses more flip-flops but typically decoding logic for the next state and output are simple.

Can modify the reduced state table

- We can add the codes for the states and depict the state table.
- We do this next.

Recall we had a table like this (see part 3)

	Next State		Output	
Present State	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	d	0	1
e	a	d	0	1

State table with code assignment

Reduced State Table with Binary Assignment 1

Present State	Next State $x = 0$	Next State $x = 1$	Output $x = 0$	Output $x = 1$
000	000	001	0	0
001	010	011	0	0
010	000	011	0	0
011	100	011	0	1
100	000	011	0	1

A note on terminology

- The name transition table is also given for a state table with binary assignment.

The complete design procedure

- The first three steps in the procedure for designing synchronous sequential circuits can be described as follows:

1. Derive a state diagram from specifications of the desired operation.
2. Reduce the number of states if required.
3. Assign binary values to the states (Remaining steps in next slide)

Remaining steps …

4. Obtain the binary-coded state table.
5. Choose the type of flip-flops (D/T/JK) to be used in the design.
6. Derive the flip-flop input equations and output equations.
7. Draw the logic diagram (circuit).

Note: We will also introduce something called excitation tables soon.

A Design Example

- Using the procedure described, we will design a sequence detector in the next part.
- The sequence detector is a circuit that detects a specific sequence in a string of bits (for example, three or more consecutive 1's) coming through an input line.

Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- C. Roth and L.L. Kinney, Fundamentals of Logic Design, Gamma Publications, 2021.